

Zürich, April 11, 2026

1. How does your work advance the field of green chemistry?

Our work advances green analytical chemistry by introducing a practical, user-tunable strategy that substantially reduces titrant and solvent consumption in one of the most widely used laboratory techniques, potentiometric titration, without sacrificing accuracy or precision. By replacing full-curve acquisitions with an intelligent, criteria-driven early-stop protocol based on Gran–Schwartz linearization, the method directly addresses the long-standing tension between analytical reliability and resource efficiency.

2. Please describe your specific green chemistry achievement, either quantitatively or qualitatively

The key achievement is a systematic 10–30 % reduction in titrant and solvent consumption per titration. This is accomplished by automatically determining an optimal earliest acceptable point volume using a transparent user-defined criterion (agreement with the full-dataset EQP within $\pm$ error). The strategy was validated on both TRIS/HCl and KHP/NaOH systems ($n = 6$ replicates each), delivering excellent repeatability (SD typically 0.010–0.050 mL) while avoiding unnecessary post-equivalence titrant addition. The entire workflow is implemented in the open-source Python package GranTED, making the green improvement immediately accessible to the community.

3. How could your work be made greener and be elevated by further research?

Further research could elevate the method by implementing real-time, on-the-fly adaptive termination: the same three criteria could be evaluated continuously during the titration, allowing automatic, sample-driven stopping as soon as the Gran–Schwartz fit satisfies the user thresholds. This would eliminate the need for a separate method-development titration in many cases. Additional improvements could include support for complexometric, precipitation, and redox titrations, as well as the analysis of polyprotic substances for additional green analytical workflows.